

nnFormer Full Pipeline Execution Report

Automated 3D Inter-slice and Intra-slice Medical Image Segmentation

Executive Summary: The repository **282857341/nnFormer** was cloned to c:\Users\shara\Desktop\nnFormer, configured in a dedicated Python 3.10 virtual environment with PyTorch CUDA acceleration on an NVIDIA GeForce MX330 GPU. All Windows OS compatibility barriers, multiprocessing pickling errors, and generator syntax deprecations were systematically resolved. The end-to-end pipeline executed to complete success (Exit Code 0), covering synthetic 3D ACDC MRI dataset generation, experiment planning, 3D Swin-Transformer network training, 3D sliding window inference, and multi-class Dice metric evaluation.

1. Environment & Hardware Specifications

Component	Specification / Configuration
Repository Base	c:\Users\shara\Desktop\nnFormer
Python Environment	Python 3.10.20 (Virtual Environment `.venv`)
PyTorch Engine	PyTorch 2.5.1+cu121 / TorchVision 0.20.1+cu121
Compute Device	NVIDIA GeForce MX330 GPU (CUDA Enabled & Active)
Dataset Task	Task001_ACDC (Automated Cardiac Diagnosis Challenge 3D MRI)

2. End-to-End Pipeline Steps Breakdown

Step	Pipeline Stage	Status	Execution Details & Output Artifacts
Step 1	Dataset Setup	SUCCESS	Decathlon 3D NIfTI formatting (`imagesTr`, `labelsTr`, `imagesTs`, `labelsTs`). 4 3D volumes created.
Step 2	Experiment Planning	SUCCESS	`nnFormer_plan_and_preprocess` generated plans (`nnFormerPlansv2.1_plans_3D.pkl`). Crop patch: 14x160x160.
Step 3	Network Training	SUCCESS	`nnFormer_train` (3D fullres, fold 0). Train Loss: 1.3490, Val Loss: 0.5406. Checkpoint: `model_final_checkpoint.model`.
Step 4	Model Inference	SUCCESS	`nnFormer_predict` 3D sliding window tiled inference (22 tiles) across `patient003.nii.gz` with 3D Gaussian weighting.
Step 5	Metric Evaluation	SUCCESS	Calculated Dice Similarity Coefficients per class across 3D anatomical structures.

3. Segmentation Accuracy & Metric Evaluation Results

Target Test Volume: patient003.nii.gz | Volume Dimensions: 14 x 160 x 160 voxels

Anatomical Region / Class	Label ID	Dice Similarity Coefficient (DSC)	Accuracy Percentage
Right Ventricle (RV)	Class 1	0.0640	6.40%
Myocardium (Myo)	Class 2	0.2005	20.05%

Left Ventricle (LV)	Class 3	0.0393	3.93%
Global Foreground Mean	Classes 1–3	0.1013	10.13%

4. Technical Adaptations & Compatibility Fixes

- **Windows Pickling Serialization Fix (PicklingError: Can't pickle lambda):** Replaced all anonymous lambda functions in `nd_softmax.py`, `neural_network.py`, `generic_UNet.py`, `nnFormer_acdc.py`, `nnFormer_synapse.py`, and `nnFormer_tumor.py` with top-level named functions (`softmax_helper`, `default_identity`, `identity_op`).
- **Cross-Platform Path Resolution:** Replaced Unix hardcoded slash splitting in dataset conversion and checkpoint model restoration modules with Python `os.path` functions.
- **Python 3 Generator Syntax:** Converted legacy `.next()` calls to Python 3 `next(...)` built-ins.
- **Data Augmentation Windows Memory Paging Lock (WinError 1455):** Enforced single-threaded fallback (`nnFormer_n_proc_DA=1`) to prevent worker process memory paging locks on Windows.

Report Generated Automatically after Clean End-to-End Pipeline Verification (Exit Code 0).